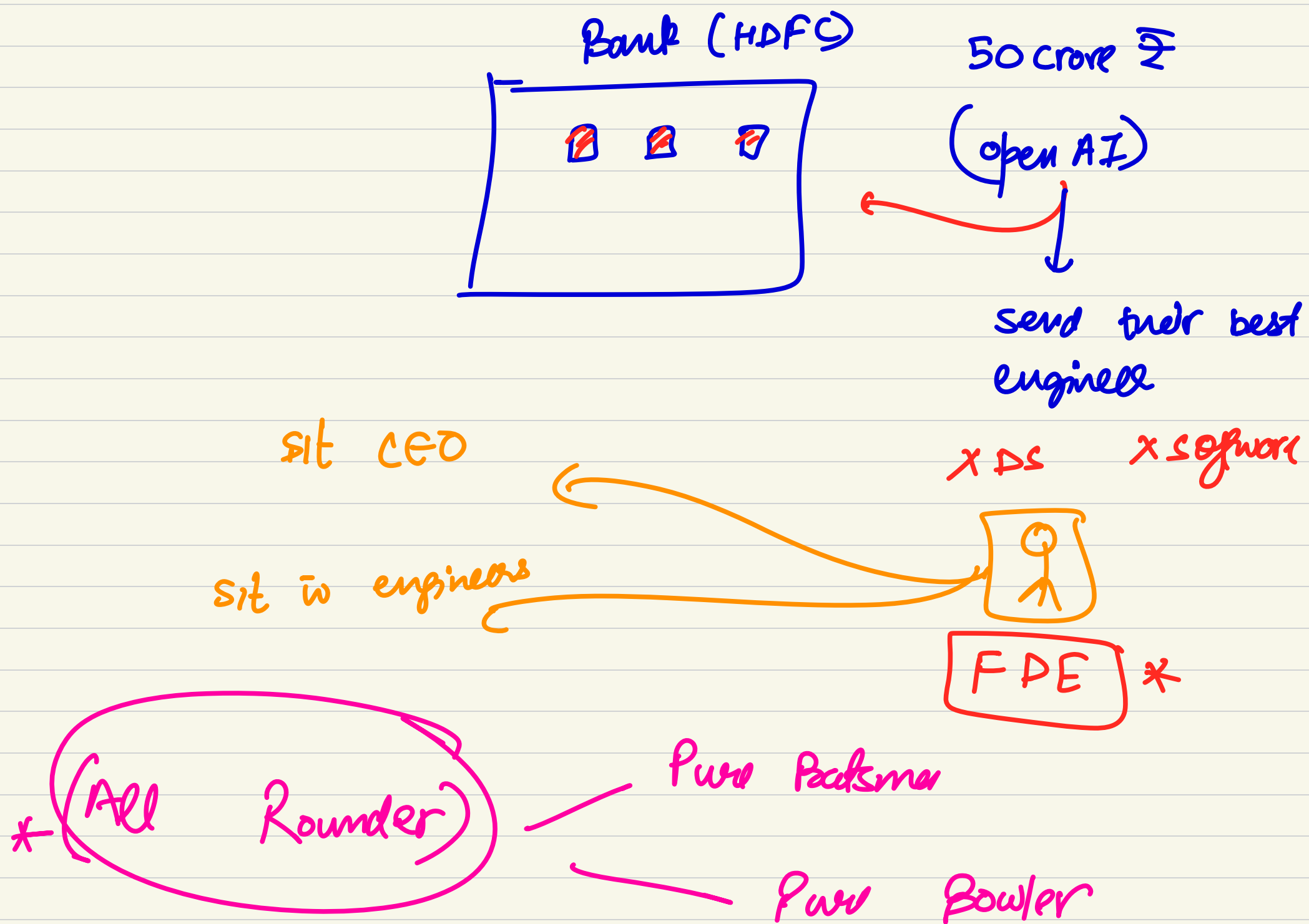


Roadmap to Become - FDE



History

Agenda

FDE (DS vs ML Engineer)

→ A day in the life of

→ [Custom Roadmap]

≡

① Visual → LLM work
→ RAG

→ Real Case Study
(Databricks)

→ some free resources
(FDE)

F

D

E

Forward

Deployed

Engineer

Dimension	Data Scientist	ML Engineer vs FDE
Primary Focus	Statistical modeling and past patterns	Production infrastructure and MLOps pipelines
Works With	Datasets, Jupyter notebooks, experiment logs	Kubernetes, CI/CD, model serving infrastructure
Delivers	Accuracy metrics and model performance reports	Scalable, reliable ML systems in production
Core Skills	Statistics, Python, classical ML algorithms	DevOps, distributed systems, cloud platforms
Salary in India	17 to 25 LPA at senior levels	18 to 25 LPA at senior levels
Best Analogy	The research doctor in a hospital lab	The hospital's infrastructure engineer

(2-5 yrs of exp)

[25 to 45 LPA]

FDE

Day in the - FDE

(Databricks)

FDE

Aman

Large

NBFC
client

Their RAG-based loan document
analysis system is producing

9:00 am : (ask structured ques)

10: am : open codebase (1.5 hours)
Naive chunking (Problem)

11:30 → update code → deployment

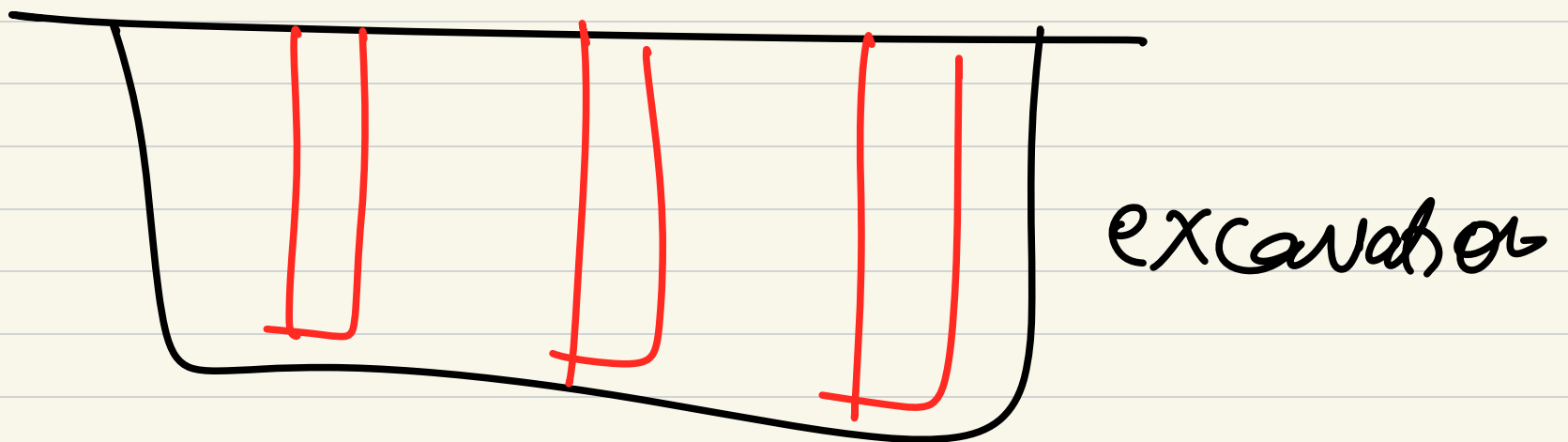
2:00 → done w changes

4:00 → document in confluence

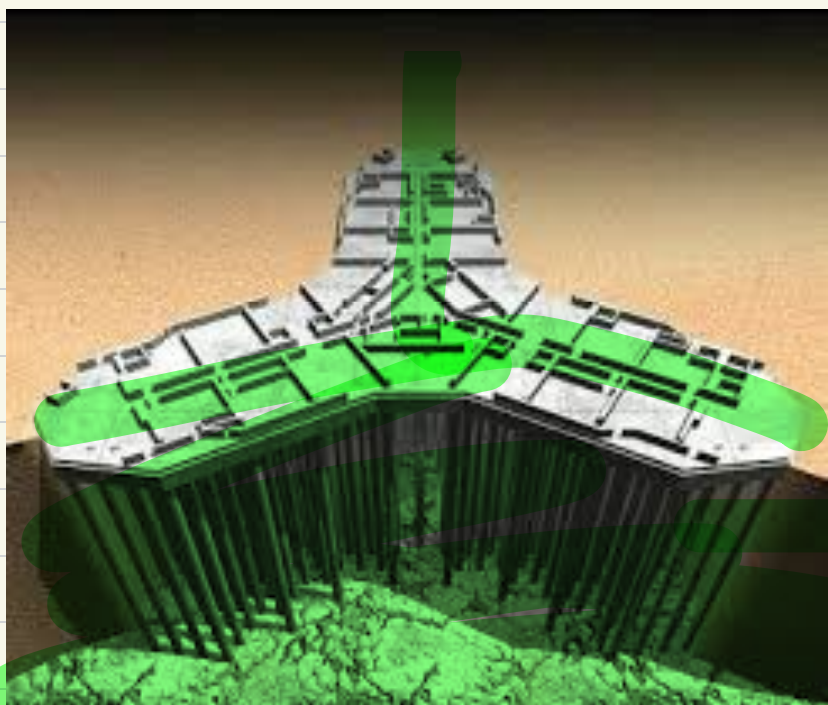
67% — 91%

5:00 → join prod standup

How to become a FDE



Buoy Khalifa



(v. strong)

Layer 1

Layer 1 — The Foundation (Month 1 to 2)

Core Python and Classical ML

- Python for data science: NumPy, Pandas, Matplotlib, Scikit-learn
- Classical ML: Linear Regression, Logistic Regression, Decision Trees, Random Forests, Gradient Boosting
- Evaluation metrics: Precision, Recall, F1 Score, ROC-AUC — understand the business meaning, not just the formula
- Statistics fundamentals: probability distributions, hypothesis testing, confidence intervals

most overlooked topic

2 months ago Google AI solⁿ architected
* [Derive central Limit Theorem]

Layer 2 The Core AI Stack

Deep Learning and Neural Networks

- Feedforward networks, backpropagation, activation functions
- Convolutional networks for vision context; RNNs and LSTMs for sequence context
- Transformers: attention mechanism, positional encoding, encoder-decoder architecture — this is the engine inside every LLM

Transformer Architecture

2017 *

Large Language Models

(LLMs)

- How LLMs are trained: pre-training on internet-scale data, supervised fine-tuning, RLHF
- Key parameters: tokenization, context windows, temperature, top-p, top-k
- Prompt engineering: zero-shot, few-shot, chain-of-thought, system prompts
- Major model families: GPT-4o, Claude 3.5 Sonnet, Gemini 1.5 Pro, Llama 3, Mistral

7 *

Embeddings and Semantic Search

- What embeddings are: dense vector representations of meaning, not just keywords

Layer 3 — The FDE Core: RAG and Agents (Month 4 to 6)

[FDE specific]

Retrieval-Augmented Generation

- The RAG progression: Naive RAG to Advanced RAG to Modular RAG
- Chunking strategies: fixed-size, semantic chunking, parent-child retrieval, hierarchical
- Hybrid search: dense vector search plus sparse BM25, combined with RRF fusion
- Reranking: cross-encoder models like BGE-Reranker and Cohere Rerank
- RAG evaluation with RAGAS: faithfulness, answer relevance, context precision, context recall

AI Agents and Agentic Systems

- The ReAct pattern: Reasoning and Acting in an alternating loop
- Tool calling and function calling (OpenAI and Anthropic APIs)
- LangChain: chains, agents, tool definitions, conversation memory
- LangGraph: stateful multi-step agents using directed graph orchestration
- Multi-agent design patterns: orchestrator-worker, parallel extractors, debate and verification

Layer 4 — The Engineering Layer (Month 5 to 7)

LLMOps and Deployment

- Fast API and Flask: building production REST APIs over AI systems
- Docker and containerization for reproducible AI deployments
- Cloud AI platforms: AWS Bedrock, Azure OpenAI Service, GCP Vertex AI
- Observability and tracing: LangSmith, LangFuse, Arize — monitor latency, cost, and quality
- Evaluation pipelines: automated metrics plus human-in-the-loop review queues

Data Engineering Essentials

- SQL fluency: JOINS, window functions, CTEs, query optimization
- PySpark fundamentals for enterprise-scale data processing
- ETL pipeline design: ingestion, transformation, loading, and freshness management

Layer 5 — The FDE Differentiator (Month 7 to 9)

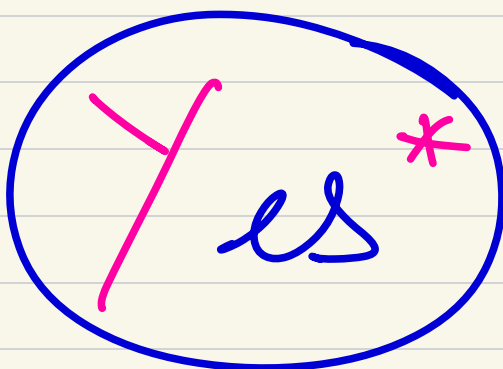
AI System Design

- End-to-end RAG system design from whiteboard to production (the case studies we cover today)
- Multi-agent financial analysis systems with ensemble verification
- Document intelligence pipelines with OCR, structured extraction, and validation
- Scalability patterns: semantic caching, vector database sharding, horizontal scaling
- Cost optimization: model routing, batch processing, tiered infrastructure

Client Communication and Consulting Skills

- How to run a discovery call: listen first, identify root causes, resist the urge to jump to solutions
- How to write a technical architecture proposal a non-engineer can understand
- How to present a failure post-mortem to a C-suite without losing trust
- Storytelling with data: building live demos in Streamlit that make AI tangible

Can all these things be self learned?

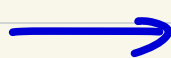


→ make sure that
you build

#

(8-10) x of time

8 months



(4-5 years)

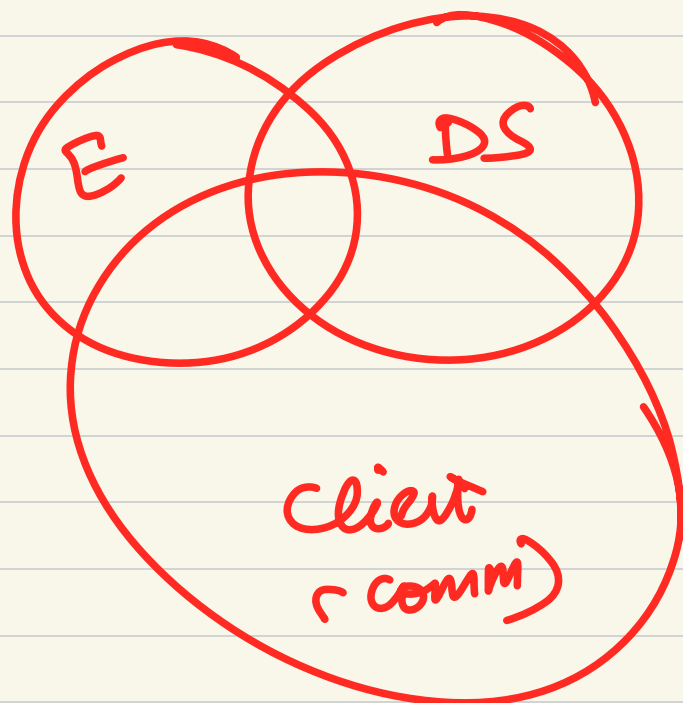
#

peer group



#

Some self
learn
resources



* (LLM → fixed knowledge) [Problem]

→ Hallucinate

RAG



This is not a RAG
class

(FDE role)

RAG

LLM



Knowledge
Source

Q and A

query



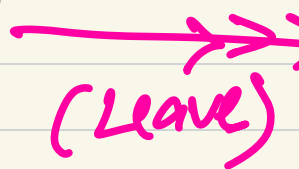
LLM



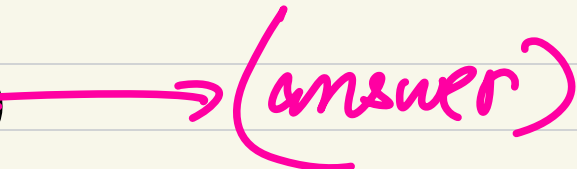
answer

RAG
Basic

query

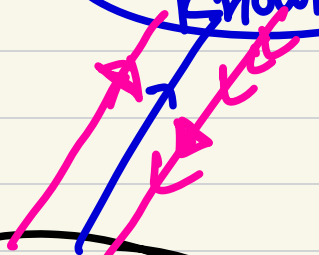


LLM

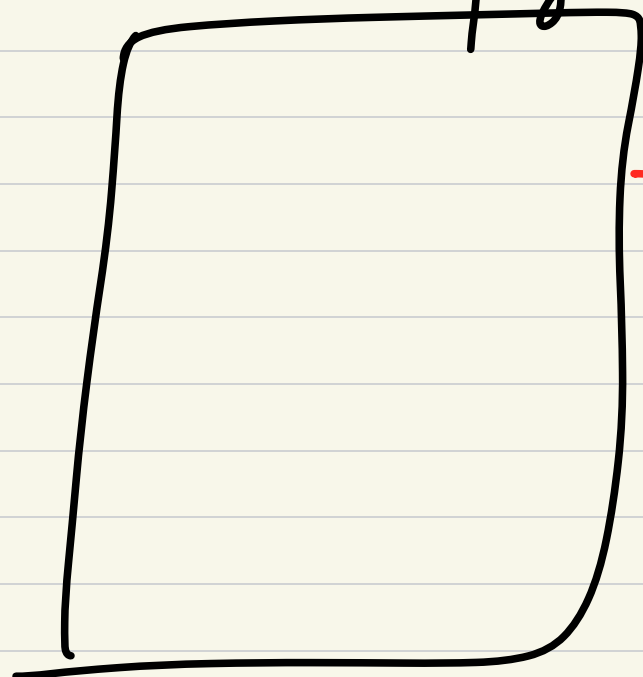


(answer)

External
Knowledge



500 page document (10GB)



→ company

Break it
Small chunks

FDE



System Design

Problem Statement

Scenario

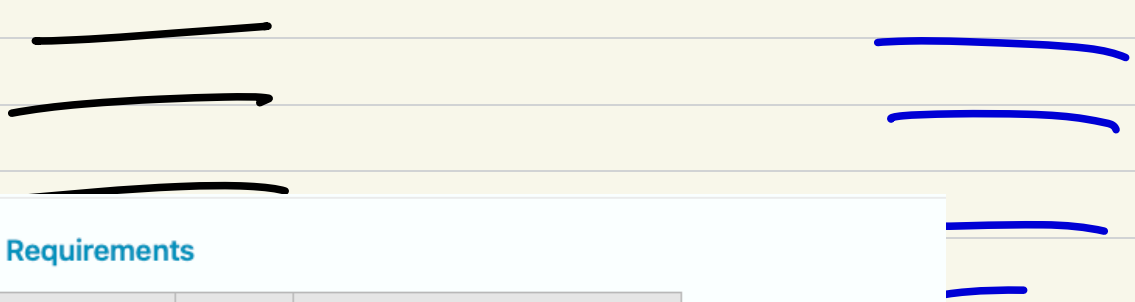
A financial services company wants to build an AI-powered search system for their internal documentation:

- 500,000 documents (policies, procedures, research reports)
- 5,000 employees across multiple departments
- Documents updated daily
- Strict compliance and audit requirements
- Need to answer questions with cited sources

① (Clarifying Question)

- Employees spend 2+ hours/day searching for information
- Keyword search returns too many irrelevant results
- Knowledge is siloed across departments
- New employees take months to become productive

② Functional & Non Functional



Functional Requirements

Requirement	Priority	Notes
Natural language Q&A	P0	Core feature
Source citations	P0	Compliance requirement
Multi-document reasoning	P1	Connect information across docs
Follow-up questions	P1	Conversational context
Document summarization	P2	Quick overview of long docs

Non-Functional Requirements

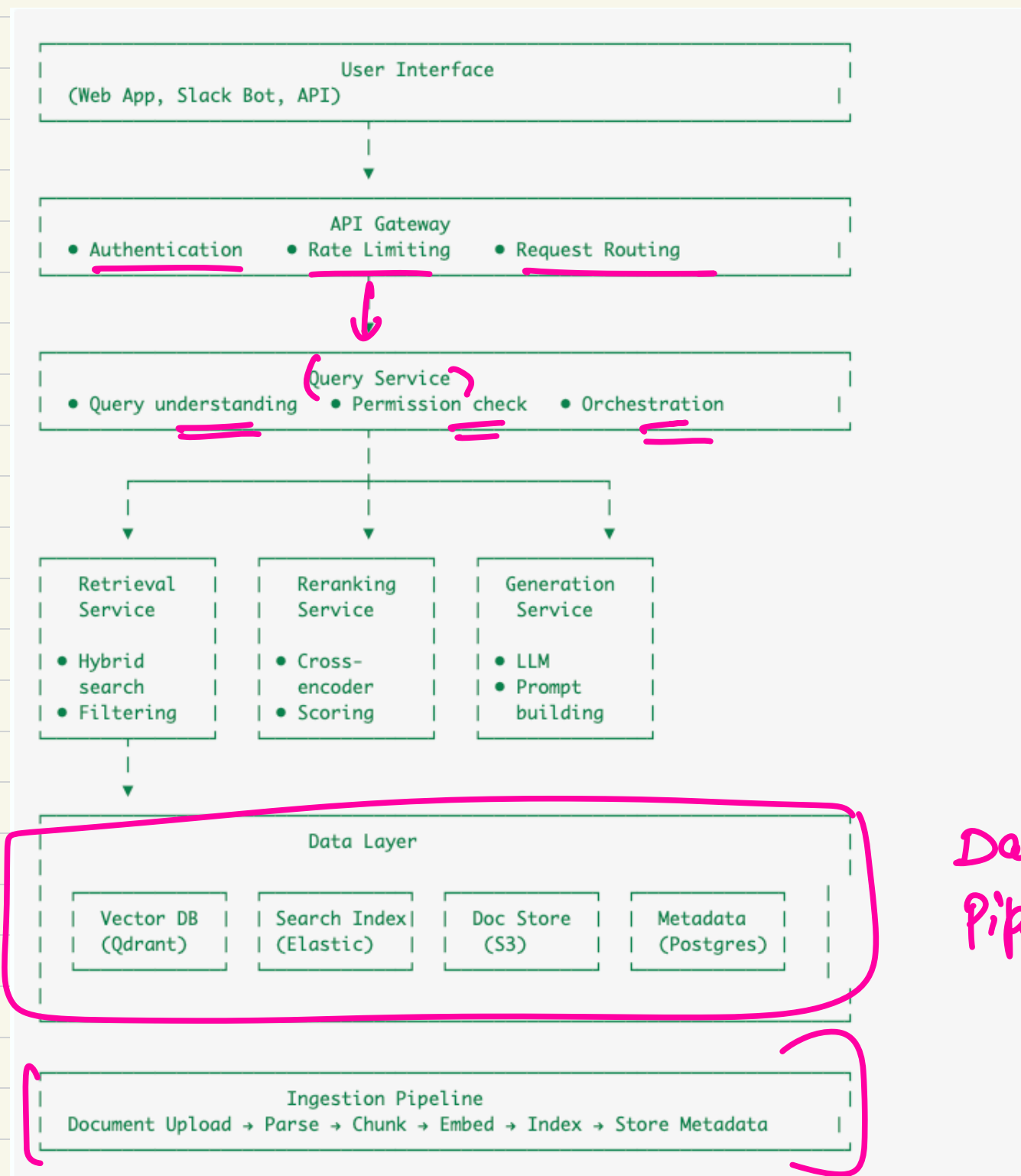
Requirement	Target	Rationale
Latency (P95)	< 5 seconds	User experience
Accuracy	> 90%	Trust and adoption
Availability	99.9%	Business critical
Concurrent users	500	Peak usage
Document freshness	< 1 hour	Policy updates

Security Requirements

- Role-based access control (RBAC)
- Audit logging of all queries
- No data leaves company network
- PII detection and handling

System Architecture

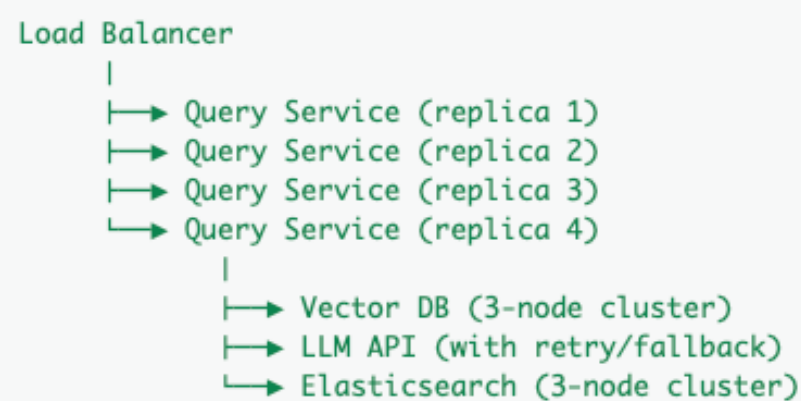
(FDE)

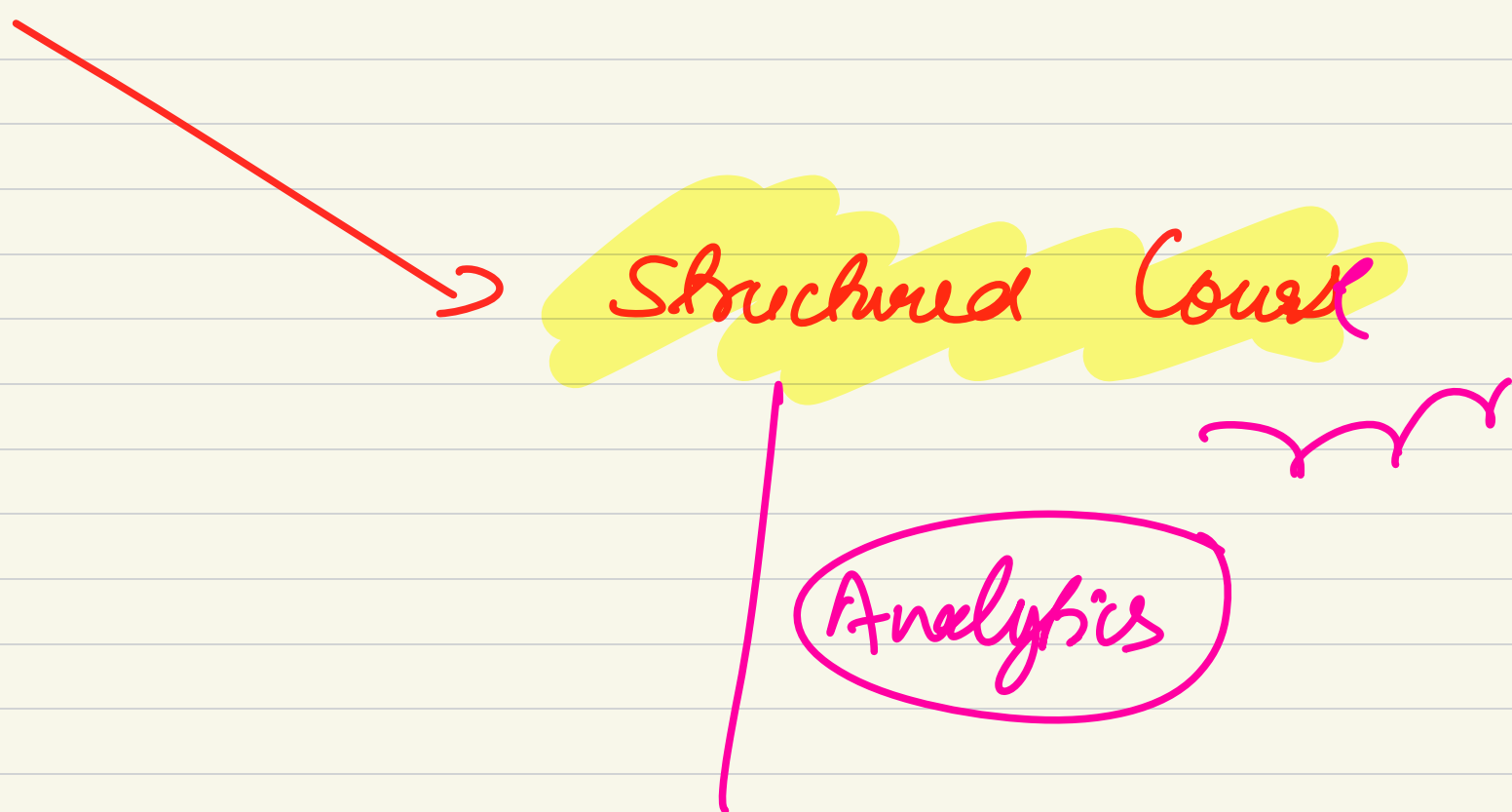
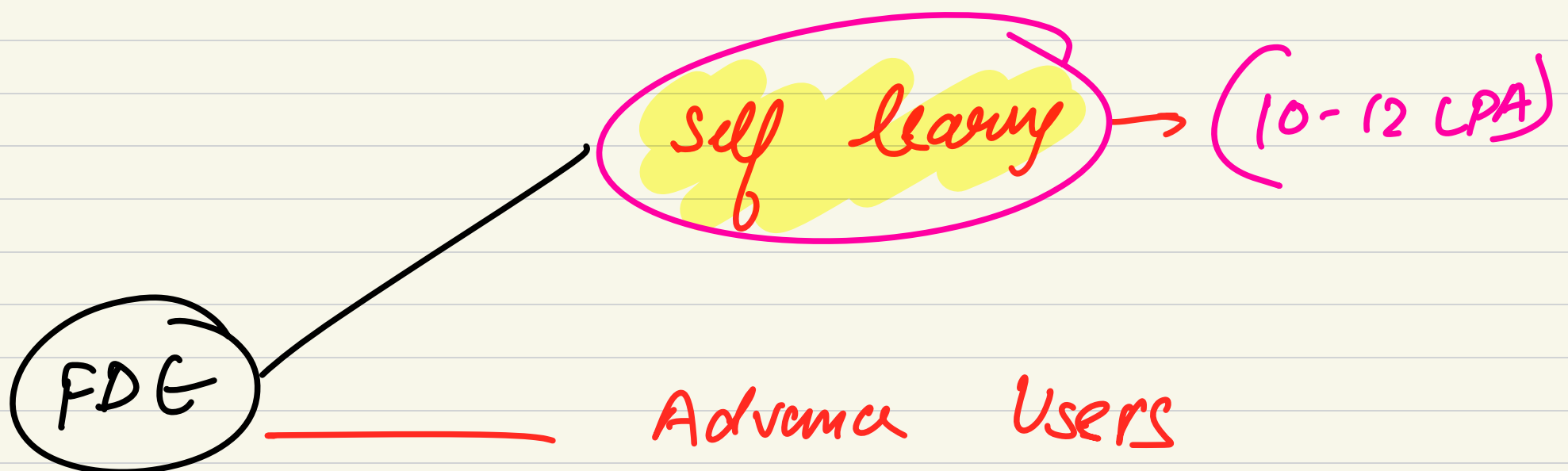


Data Pipeline

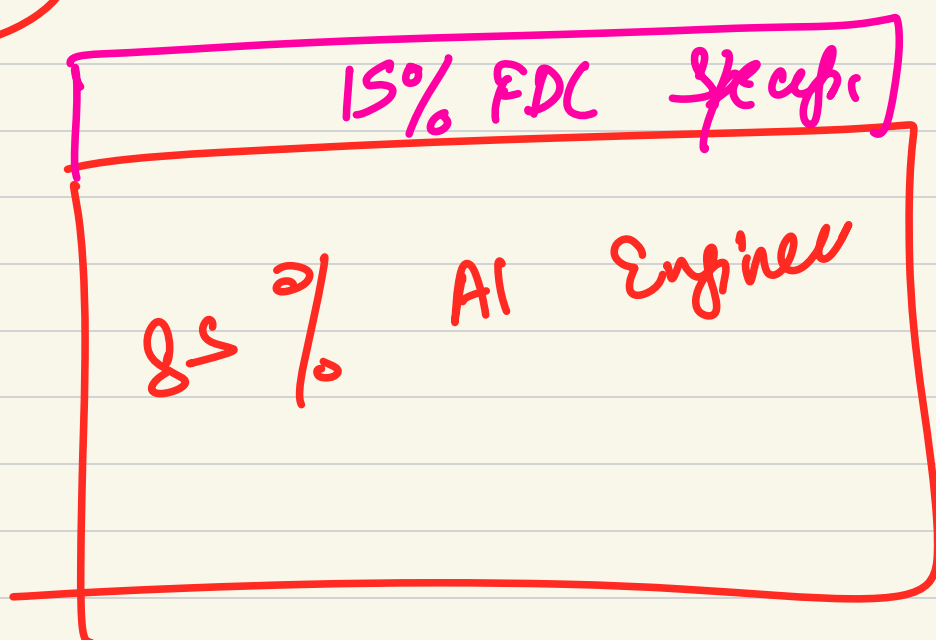
500 users

Handling 500 Concurrent Users



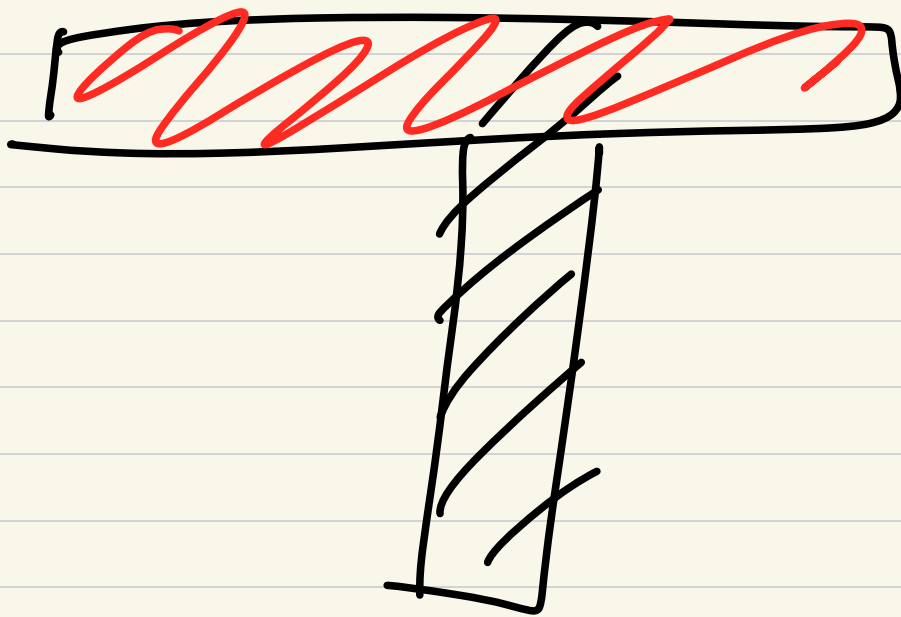


V. New



(what interviewers look for)

Dimension	What It Means in Practice
Deep Expertise (the vertical bar)	You have one area of genuine depth: <u>RAG system design</u> , or <u>agentic AI</u> , or <u>LLMOps</u> , or multimodal AI. You can go <u>three levels deeper than the interviewer on this topic without hesitation.</u>
Broad Familiarity (the horizontal bar)	You know enough about every adjacent area — <u>classical ML</u> , <u>cloud platforms</u> , <u>data engineering</u> , <u>product thinking</u> — to have intelligent conversations and avoid making embarrassing gaps visible.
Communication Clarity	You can explain a complex architecture decision to a software engineer in technical <u>terms</u> , and explain the same decision to a CFO in business terms, without losing accuracy in either direction.
Intellectual Honesty	You admit what you do not know. You follow an I-do-not-know statement with here-is-how-I-would-find-out. Interviewers are evaluating whether they can trust your judgment in front of their customers.



Salaries

Junior FDE
(28 - 35 LPA)
freshers

45 - 70 LPA (india)

200k \$ + S.O → 200k \$

AWS / Fast API

Timeframe	Market Outlook
2025 to 2027	Every Fortune 500 company builds an internal AI team. FDE demand doubles. Candidates with production experience are hired before graduation in many cases.
2027 to 2030	Agentic AI becomes mainstream across enterprises. FDEs who understand multi-agent system design will command the highest compensation in the market.
India-specific	The Indian AI market is growing at 25% per year. Bangalore is emerging as the primary AI engineering hub in Asia, with both product companies and global service delivery centers hiring aggressively.
Career ceiling	The FDE path leads to: AI Practice Lead, VP of AI and ML, Chief AI Officer. The professionals building these skills today will be in executive roles by 2030.

GCC

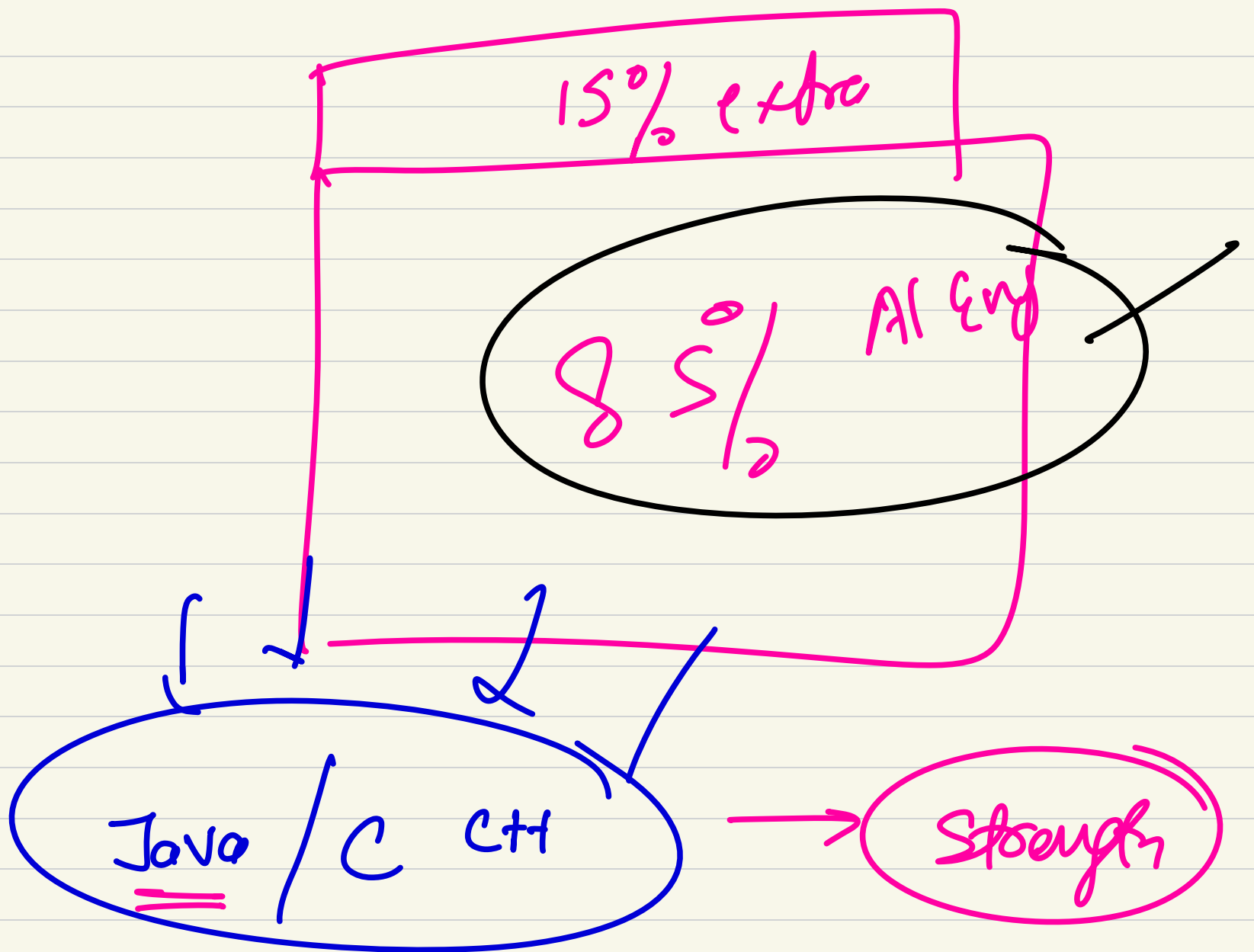
_____ X _____ X-

Doubts → Callback Poll
→ specifically

5 mins → LinkedIn "

AI Engineers → Python DS AI

Colleg → SWT



Python →

Aggr 3-4 years

LLM (MCP) Tools

HTTP

Luhedun